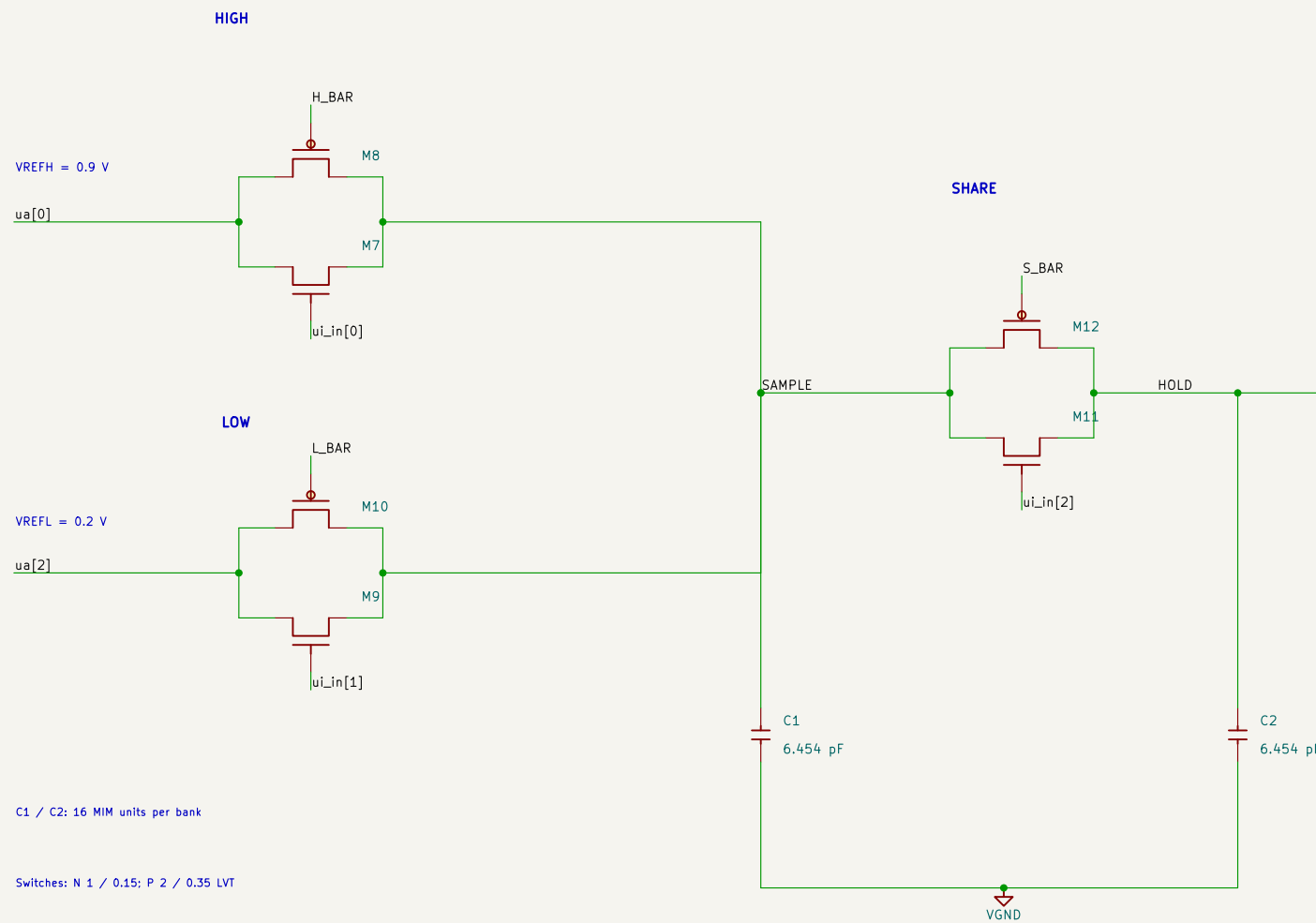


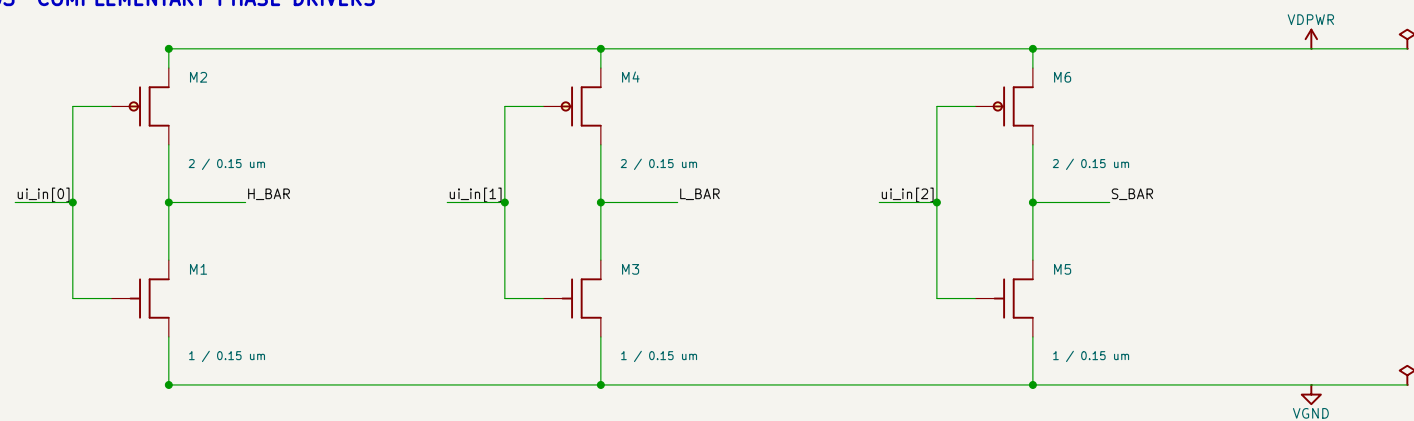
## SUAREZ SERIAL CHARGE-SHARING DAC

SKY130 | 1.8 V | 8 bits, LSB first | 161 x 225.76  $\mu\text{m}$  | W / L dimensions in  $\mu\text{m}$

## 01 REFERENCE SELECTION AND CHARGE SHARING



### 03 COMPLEMENTARY PHASE DRIVERS



## OPERATING SEQUENCE

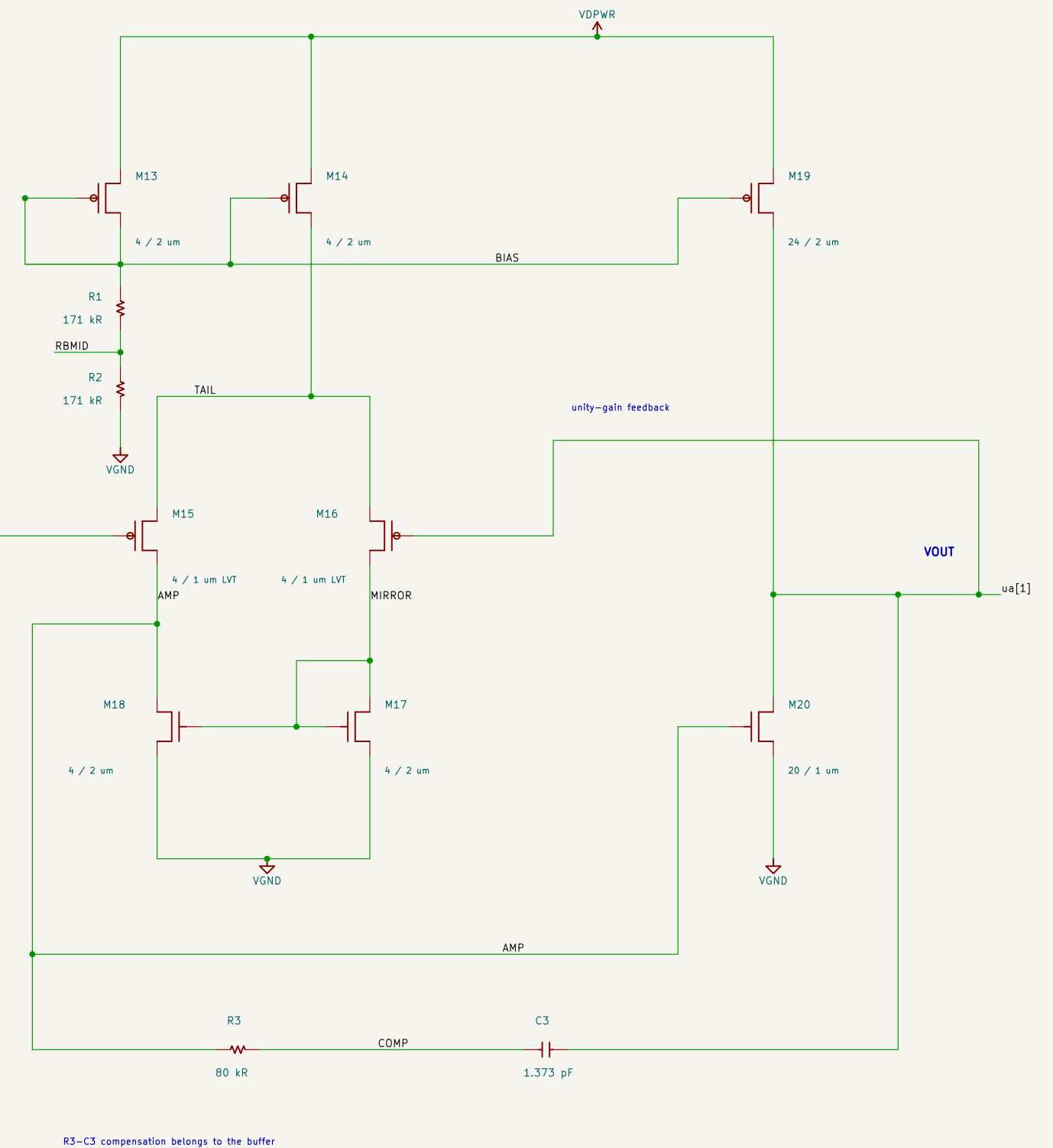
Initialize: HIGH=0, LOW=1, SHARE=1 for 8 us. Then all phases off for 0.2 us.

8 bits, LSB first: charge 2 us; dead 0.2 us; share 2 us; dead 0.2 us. Read 3 us after bit 7.

HIGH / LOW never overlap. LOW / SHARE overlap only during initialization. External controller supplies dead time.

Bodies: all NMOS  $\rightarrow$  VGND; all PMOS  $\rightarrow$  VDPWR. Body ties are explicit in SPICE / silicon.

## 02 UNITY-GAIN BUFFER / BIAS, FEEDBACK AND COMPENSATION



SKY130 / Tiny Tapeout

Sheet: /  
File: suarez\_dac.kicad\_sch

Title: Suarez two-capacitor DAC

|                     |       |
|---------------------|-------|
| Size: A3            | Date: |
| KiCad E.D.A. 10.0.6 |       |

Rev: 0.3  
Id: 1/1